Determination of the Boltzmann constant from thermal noise measurements

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Abstract

The aim of this lab experience is to determine the Boltzmann constant k_B by measuring the thermal noise power across an unbiased resistor at room temperature. To investigate this tiny noise an operational amplifier is used. The first method involves examining the V_{RMS} of the noise in the time domain while the second method involves the estimation of the constant power density of the noise in the frequency domain. The values of k_B so obtained are $k_B = (1.38 \pm 0.02) \cdot 10^{-23}$ J/K with the first method and $k_B = (1.39 \pm 0.02) \cdot 10^{-23}$ J/K with the second one.

Introduction

The Johnson–Nyquist noise is a source of voltage and current noise in any circuit at finite temperature due to the thermal agitation of the charge carriers inside any conductor. This type of noise is present even when no net current is flowing in the circuit, it has a Gaussian profile and it's approximately white: it has a constant power spectral density (PSD) across the whole frequency spectrum.

This constant noise density S in the power spectrum of the Johnson noise is given by the equation:

$$S = 4k_B T \,\mathrm{Re}(Z) \tag{1}$$

where T is the temperature of the system and Z is the complex impedance of a given circuit. Only the real part is considered since the imaginary part, the reactance, does not lead to energy dissipation.

The power spectral density S is given in voltage variance per hertz of bandwidth (V^2/Hz) and, for a given equivalent bandwidth Δf, the root mean square (RMS) of the noise on the voltage is given by:

$$V_{RMS}^2 = 4k_B T \,\mathrm{Re}(Z) \Delta f \tag{2}$$

As it can be seen the Boltzmann constant k_B appears in the above formulas suggesting that through an accurate measure of the Johnson–Nyquist noise it is possible to estimate this fundamental physical constant.

A similar approach has been used to measure k_B up to a relative standard uncertainty of $2.7 \cdot 10^{-6}$ by Jifeng Qu et al. in 2017 [1].

Materials and methods

The schematics for the experimental setup can be seen in Fig.(1).

The idea behind this experiment is to measure the noise over an unbiased resistor R by properly amplifying this weak signal with an op amp.

The three key equipment parameters are the total impedance of the circuital elements before the amplifier, and the amplifier specifications: gain and bandwidth.

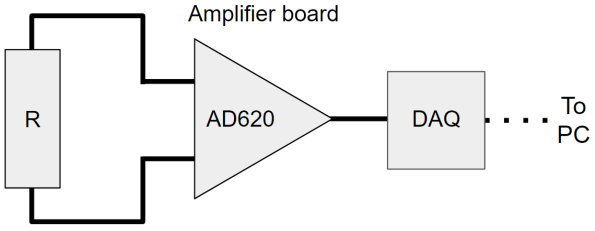


Figure 1: Experimental setup schematic

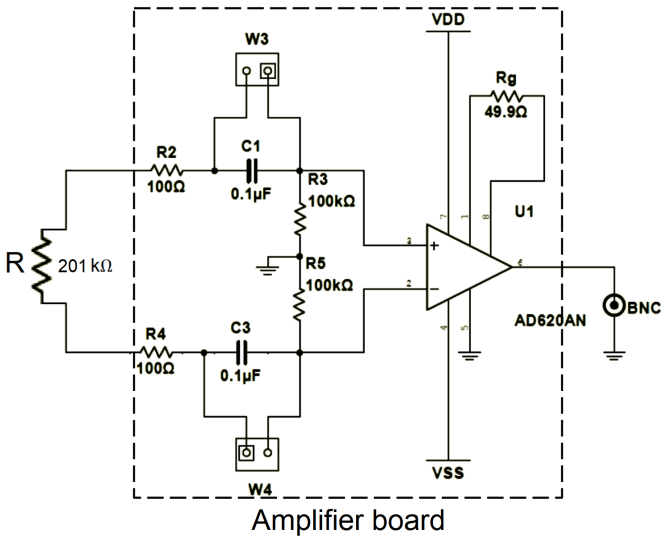


Figure 2: Amplification circuit schematics including the resistor R. The BNC output is connected to the DAQ

Instrumentation

The DAQ used is a Digilent Analog Discovery 2 which is software controlled and is connected via USB to the PC. The instrument consists in a two-channel oscilloscope with 14-bit ADC and a two-channel function generator. The function generator is used to probe the gain profile of the amplifier.

The amplifier board used in this experiment was built in-house and is based on, as shown in Fig.(2), the low noise differential amplifier integrated circuit AD620AN from Analog Devices [2]. The board is powered by two 9 V batteries in parallel and is kept inside an aluminium box to reduce external noise. This device has a nominal gain of 1000 and a nominal input noise power of nearly 10^{-12} V^2 inside its bandwidth of 10 kHz. The gain and bandwidth will also be experimentally assessed.

The board comes with a built-in high-pass filter (HPF) realized with an RC circuit (C1-R3 and C3-R5), that can be shorted and thus bypassed via the jumpers W3 and W4. The purpose of this high-pass filter, with its cutoff frequency of 16Hz, is to attenuate the 1/f noise (dominating at low frequencies) that could saturate the differential amplifier. For the whole experiment these jumpers remain un-shorted and thus the HPF remains enabled.

The gain profile $G(f)$ of the amplifier is probed, to obtain both the gain and bandwidth, by generating sine waves of fixed amplitude at logarithmically spaced frequencies below and above the nominal bandwidth of the amplifier, such that the gain is simply the ratio between the output and input signals amplitudes. The actual profile is obtained by fitting the collected data points with a Butterworth low-pass filter frequency profile:

$$G(f) = G\frac{1}{\sqrt{1+(f/f_c)^2}}$$

The measured profile can be seen in Fig.(3). The obtained amplifier gain G is 961 ± 2 and its cutoff frequency f_c is 10.92 ± 0.09 kHz.

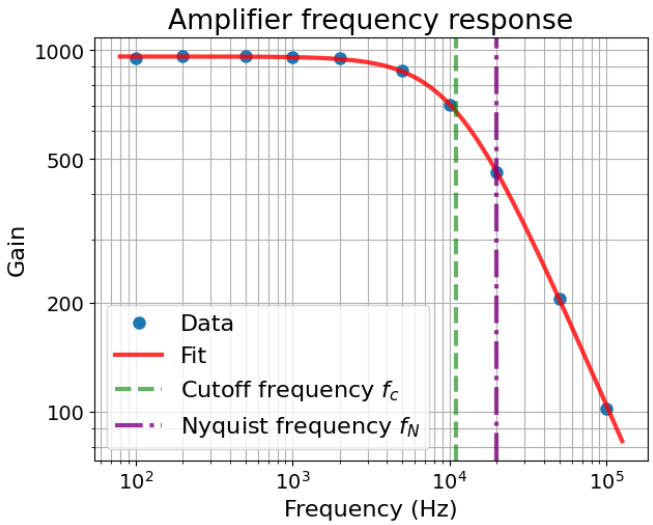


Figure 3: Gain profile of the differential amplifier

During the acquisition of the noise the DAQ was kept at sampling frequency of 40kHz, a decision that, in retrospect, is not optimal, but that it is possible to account for. The equivalent noise bandwidth Δf is normally calculated as such:

$$\Delta f = \frac{1}{G^2}\int_0^\infty G(f)^2 df = \frac{\pi}{2} f_c$$

but since the sampling frequency is so close to our f_c it influences Δf as follows. In this work the DAQ is modelled as a second low pass filter with cutoff frequency f_N at 20kHz (the Nyquist frequency).

This gives:

$$\Delta f = \frac{1}{G^2}\int_0^\infty \frac{G(f)^2}{1+(f/f_N)^2}df \approx 1.015 f_c$$

For a more accurate description of Δf, aliasing must be taken into consideration: the noise power present at frequency higher than $f_N = 20\,\text{kHz}$ is folded back into the spectrum and added (voltage gain is summed under quadrature) resulting in a fictitious rise to the gain of the setup. The outcome of this process is the "Final transfer function" shown in Fig.(4).

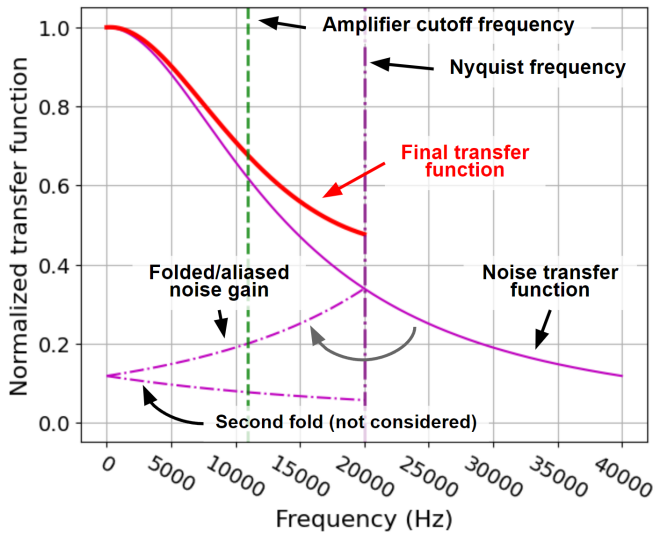


Figure 4: Final transfer function accounting for the aliasing in red. The "Noise transfer function" is the multiplication between the amplifier G(f) and the DAQ low-pass filter transfer function. Discarding the second fold is equivalent to considering a "brick-wall" filter at the 40 kHz sampling frequency

The integral of this final transfer function $F(f)$ in the relevant frequency span is:

$$\Delta f = \frac{1}{G^2}\int_0^{f_N} F(f)^2 df \approx 1.031 f_c$$

The full form of $F(f)$ is given in the Appendix.

The integration grants us an equivalent bandwidth of $\approx 11\,260 \pm 90$ Hz.

The attached resistor R has a resistance, measured with a multimeter, of $R = 201 \pm 1$ kΩ, but, as it can be seen from the schematics in Fig.(2), the total input impedance needs to be calculated from all the circuital elements on the amplifier board to the left of the op amp. The total impedance is thus limited by the resistances R3 and R5 of the high-pass filter, parallel to the input of the amplifier: the maximum achievable input impedance is as such about 200 kΩ, but this value quickly drop off at higher frequencies. Assuming all the reported values of the circuital elements on the board shown in Fig.(2) are exact, the theoretical behaviour of the total impedance is shown in Fig.(5). Ignoring the small region near the HPF cutoff the real part of the total input impedance Z is 100.3 ± 0.2kΩ. From now on the real part of the total impedance will be labelled as R.

Taking into account the presence of the HPF in the gain profile of the amplifier is of no concern since the first data point used for the $G(f)$ fit is at 100 Hz: well above the cutoff of the HPF.

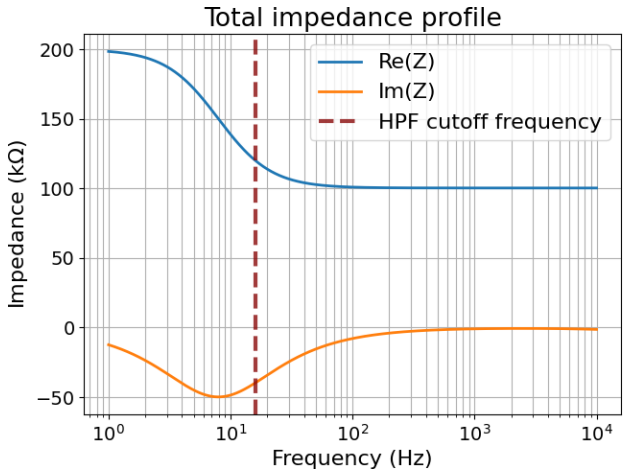


Figure 5: Real and imaginary total impedance Z

Finally the thermal noise measurement is conducted over a period of three minutes.

The temperature of the setup is not monitored and is assumed to be 20 ± 5 °C. Temperature changes during the experiment might introduce some additional unaccounted uncertainty on the obtained k_b values.

In Tab.(1) is provided a summary of the specifications of the amplifier and DAQ.

Differential amplifier AD620A [2]	
Gain G	961 ± 2
Bandwidth f_c	$10\,920 \pm 90$ Hz
Input noise density V_a	9 nV/$\sqrt{\text{Hz}}$
Input impedance	10 GΩ || 2 pF
DAQ Digilent Analog Discovery 2 [3]	
ADC bit	14
Bandwidth	30 MHz
Max sampling rate	100 MHz
Sampling rate used $(2 \cdot f_N)$	40 kHz
Setup	
Equivalent bandwidth Δf	$11\,260 \pm 90$ Hz
Total resistance R	100.3 ± 0.2 kΩ
Temperature T	20 ± 5 °C

Table 1: Brief summary of the specifications and parameters used

Method of analysis

Two methods are used for estimating k_B.

In the first method the RMS of the signal is used (Eq.(2)). Via a Gaussian fit of the noise profile in the time domain the total RMS voltage value V_{out} is obtained. It is composed by:

$$V_{out}^2 = G^2(V_J^2 + V_a^2 \Delta f)$$

where V_J is the RMS of the Johnson noise, which we are interested in, and V_a is the input noise density generated by the amplifier.

Thus V_J is given by:

$$V_J^2 = \frac{V_{out}^2}{G^2} - V_a^2 \Delta f = 4 k_B T R \Delta f$$

From here k_B can be extracted as:

$$k_B = \frac{(V_{out}/G)^2 - V_a^2 \Delta f}{4TR\Delta f}$$

In the second method Eq.(1) is used. The constant power density of the Johnson noise from the gain-normalized acquired power spectrum is given by:

$$S_{avg} = S_J + V_a \quad \text{with} \quad S_J = 4 k_B T R$$

To extract S_{avg} the mean of the PSD is calculated via an exponential fit of the power density profile distribution.

Thus k_B is simply:

$$k_B = \frac{S_{avg} - V_a}{4TR}$$

Results and discussion

In Fig.(6) the raw recording of the noise and its expected Gaussian profile are shown. The standard deviation of this profile correspond to RMS of the signal V_{out} that is evaluated to be $4.35 \pm 0.013\,\mu$V giving us a value of k_B of $(1.38 \pm 0.02) \cdot 10^{-23}$J/K.

This value is very close the real value of $k_B = 1.3806 \cdot 10^{-23}$J/K[4] showing an accuracy of this experimental method that goes beyond what was expected.

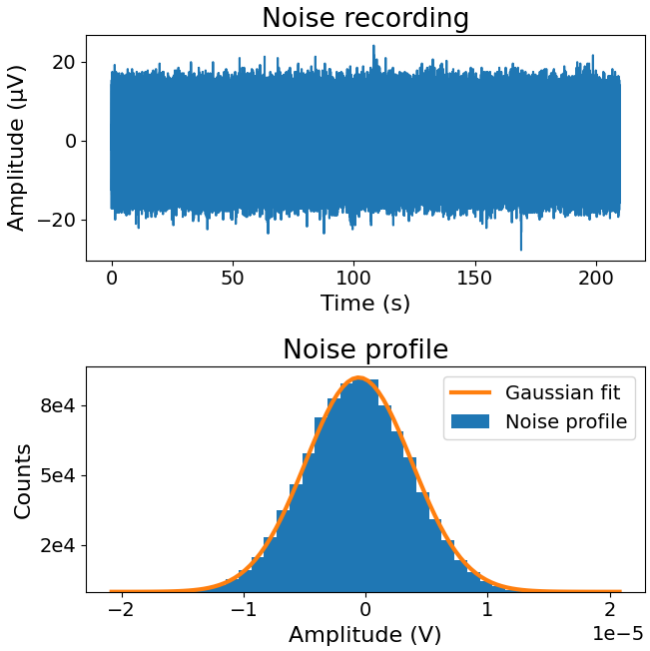


Figure 6: Raw noise recording and its Gaussian profile

To obtain the power per unit of frequency (power spectral density) of this signal a discrete Fourier transform is used. The raw Fourier transform of the collected noise along with two differently normalized spectra is shown in Fig.(7). The smoothed PSD (using a 100pts rolling mean) is shown only for illustrative purposes. It can be seen that the normalized power spectral density, for the exception of some low-frequency 1/f noise and some stray signals, is constant, as expected.

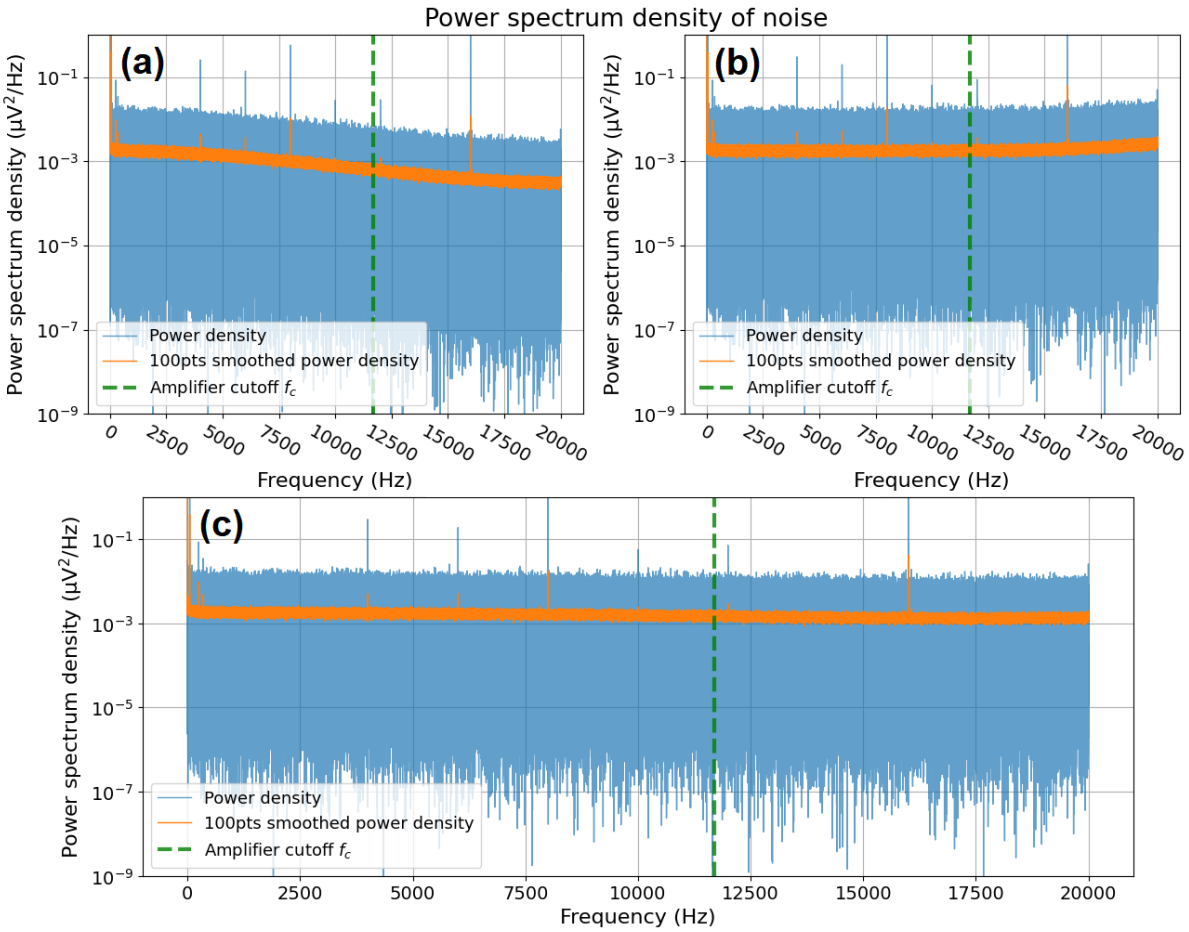


Figure 7: PSD of the acquired noise. In figure **(a)** is displayed the raw spectra without any normalization. In figure **(b)** the spectrum is normalized without accounting for aliasing: notice the higher density at high frequencies. Finally in figure **(c)** the aliasing is accounted for by normalizing the spectrum using the $F(f)$ transfer function: a very flat spectrum over all frequency

As it can be seen in Fig.(8) the profile of the PSD follows an exponential distribution and so its mean value, along with its error, can be extracted by linearly fitting the log of the profile as shown in the same Fig.(8). The R^2 of this fit is 0.998 and the PSD obtained this way has a value of $(1.716 \pm 0.002) \cdot 10^{-3}\,\mu V^2/Hz$ giving us $k_B = (1.39 \pm 0.02) \cdot 10^{-23}\,J/K$.

Conclusions

In this lab experience the Boltzmann constant is determined by measuring the thermal noise power across an unbiased resistor at room temperature. The values of k_B obtained are $k_B = (1.38 \pm 0.02) \cdot 10^{-23}\,J/K$ with the RMS and $k_B = (1.39 \pm 0.02) \cdot 10^{-23}\,J/K$ with the PSD method. Both methods are accurate and give compatible results with the real value of k_B.

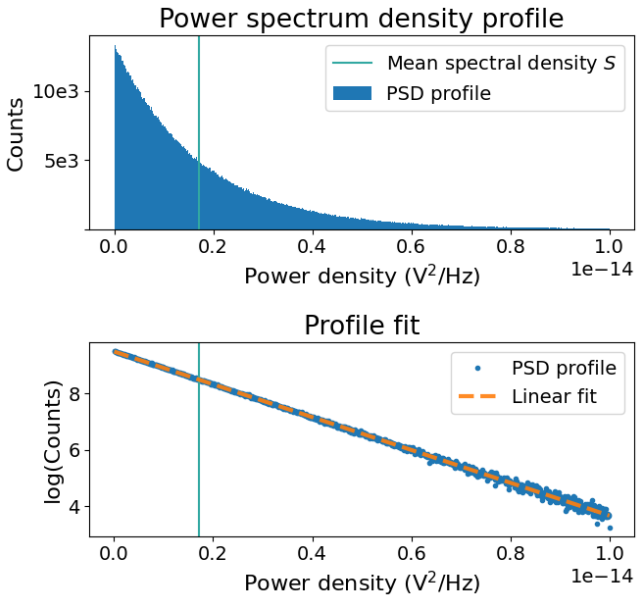


Figure 8: PSD exponential profile and its fit

References

[1] Jifeng Qu et al. "An improved electronic determination of the Boltzmann constant by Johnson noise thermometry". In: *Metrologia* 54.4 (July 2017), pp. 549–558. DOI: 10.1088/1681-7575/aa781e.

[2] Analog Devices low noise amplification IC AD620AN. URL: https://www.analog.com/en/products/ad620.html.

[3] Digilent Analog Discovery 2. URL: https://digilent.com/reference/test-and-measurement/analog-discovery-2/specifications.

[4] 2019 redefinition of SI base units. URL: https://www.nist.gov/si-redefinition/meet-constants.

Appendix

Final transfer function expression

The Butterworth low-pass frequency response is:

$$B_{F_0}(f) = \frac{1}{\sqrt{1 + (f/F_0)^2}}$$

where F_0 is the cutoff frequency of the low-pass. Using this notation $G(f)$ can be written as $G(f) = GB_{f_c}(f)$.

The "Final trasnfer fucntion" $F(f)$ has thus this expression:

$$F(f) = \frac{G}{K}\sqrt{B_{f_c}(f)^2 + B_{f_c}(-f + 2f_N)^2} \cdot \sqrt{B_{f_N}(f)^2 + B_{f_N}(-f + 2f_N)^2}$$

where K is the normalization coefficient which is equal to value of $F(f = 0)$. $B_{f_c}(-f + 2f_N)$ is the part of the transfer function that is folded back into the observed spectrum $[0,f_N]$ to take into account aliasing. If $f_N \gg f_c$ then F(f) approaches $B_{f_c}(f)$ as expected.